



AWS Cloud Development Kit (CDK)

Introduction

Imagine defining AWS infrastructure using **modern programming languages** instead of YAML/JSON.

- How do you simplify CloudFormation templates?
- How can developers write infrastructure as code using Python or JavaScript?
- How do you manage AWS infrastructure programmatically?

AWS CDK (Cloud Development Kit) allows developers to define AWS infrastructure using programming languages!

Section 1: Learn

What is AWS CDK?

AWS Cloud Development Kit (CDK) is a framework that enables defining AWS infrastructure using languages like Python, TypeScript, and JavaScript.

Why Use AWS CDK?

Traditional CloudFormation	AWS CDK
Uses JSON/YAML templates	Uses programming languages
Manual updates required	Code-driven infrastructure



Traditional CloudFormation	AWS CDK
Difficult to modularize	Reusable components (constructs)

Example AWS CDK Code

```
import * as cdk from 'aws-cdk-lib';
import * as s3 from 'aws-cdk-lib/aws-s3';

class MyStack extends cdk.Stack {
  constructor(scope: cdk.App, id: string) {
    super(scope, id);

    new s3.Bucket(this, 'MyBucket', {
      versioned: true
    });
  }
}

const app = new cdk.App();
new MyStack(app, 'MyCDKStack');
```

Why is this useful?

- Simplifies infrastructure management using code.
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Conclusion

AWS CDK modernizes infrastructure deployment using familiar programming languages.